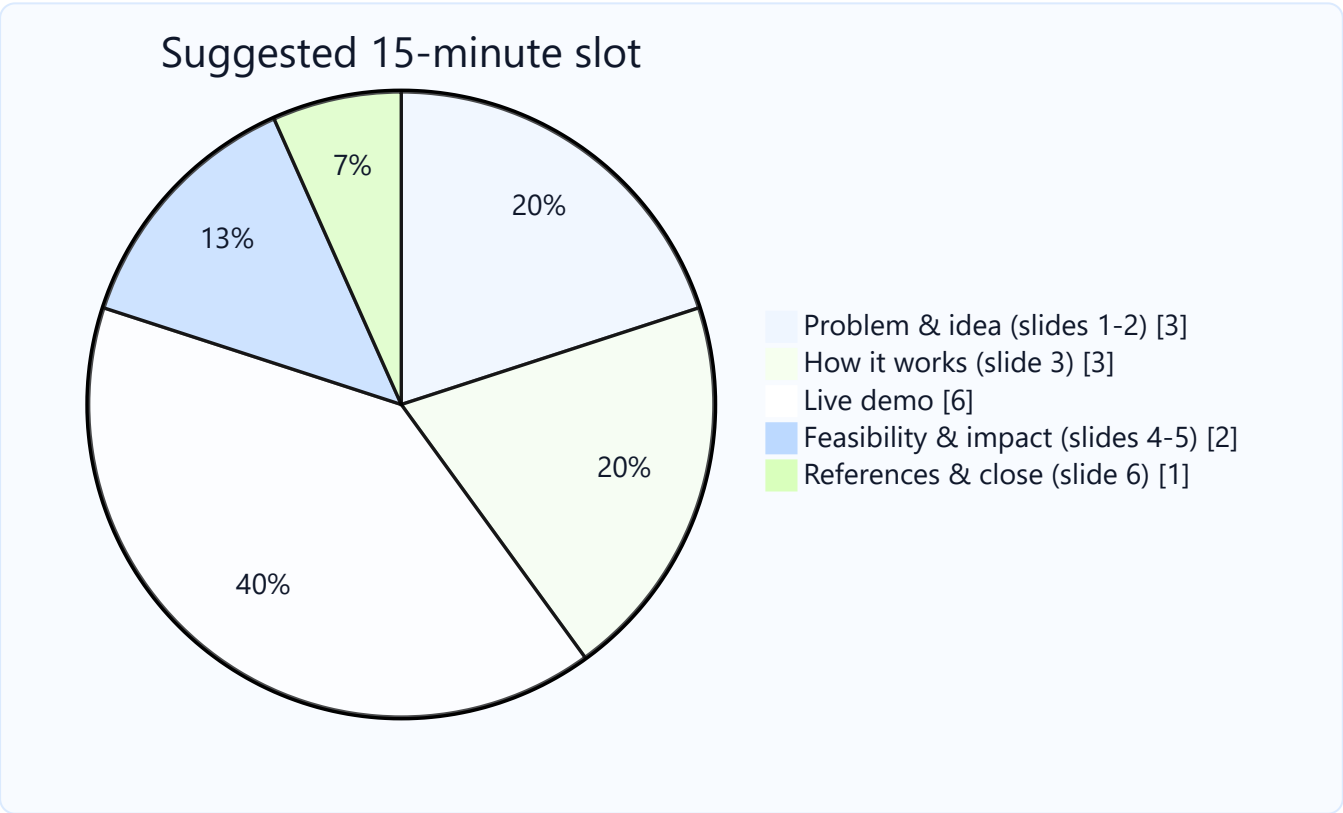


Drishti — Demo Script & Explainer

SIH 2026 · PS SIH26085 · Team LLMao Use this document to present, demo and defend Drishti. Every number in it comes from the model runs in this repository. Lines in "Say" blocks are meant to be spoken; everything else is for you.

How to use this document

If you have...	Use
2 minutes	§1 The 60-second pitch + the loop diagram
7 minutes (slides)	§5 Slide-by-slide talk track
10 minutes (live)	§4 Live demo walkthrough
Judges' questions	§7 Q&A bank
A technical reviewer	§6 Technical deep dive



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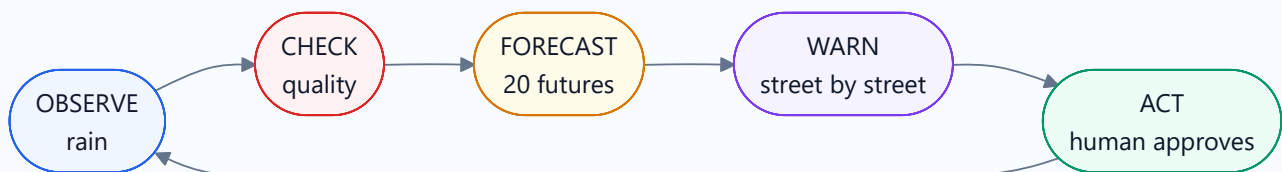
1. [The 60-second pitch](#)
 2. [The problem, explained](#)
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1. The 60-second pitch

Say: "Every monsoon, Indian cities flood in minutes. The warnings people get are city-wide — *heavy rain expected*. But a traffic officer, an ambulance dispatcher or a drainage engineer needs to know **which street, when, how deep, why**, and **what to do**."

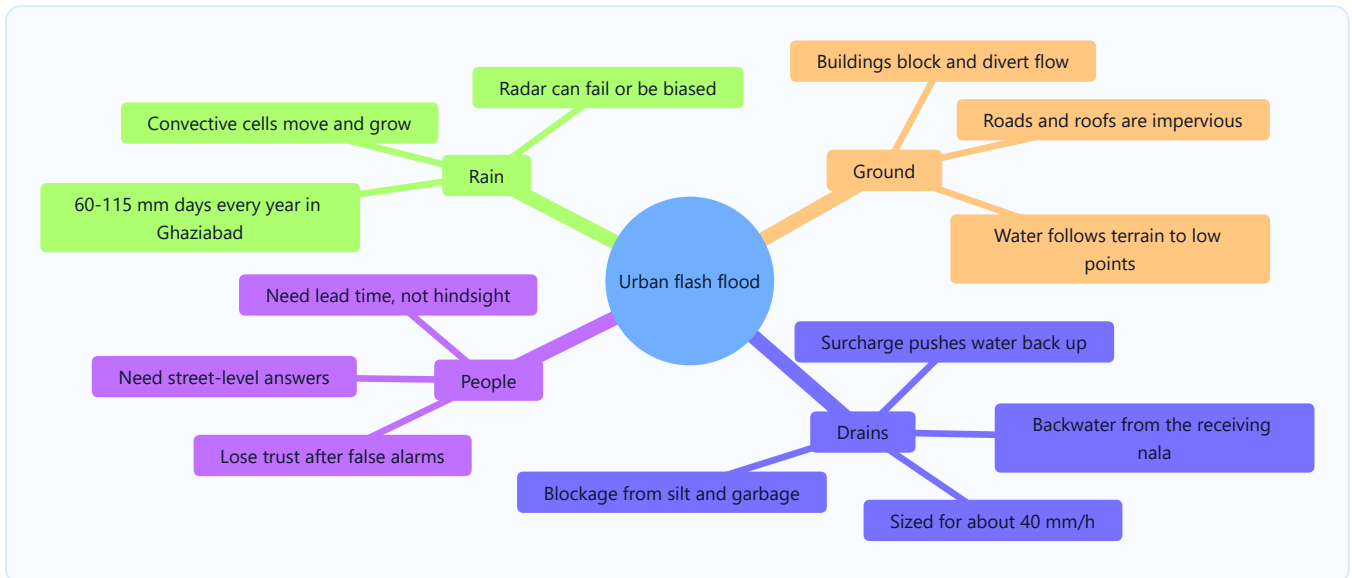
Drishti is a digital twin of a campus's streets and drains. Every five minutes it takes the rain it can see, forecasts twenty possible rain futures for the next three hours, runs them through a physics model of how water flows over the ground and through the pipes, and turns that into street-level flood forecasts — with a confidence band, a reason, and a recommended action.

It runs a full cycle in about 19 seconds on a laptop CPU. In replayed storms it warned 67% of the cells that later flooded, with 2% false alarms. And when the radar fails, it doesn't guess silently — it falls back to gauges and says DEGRADED on screen."



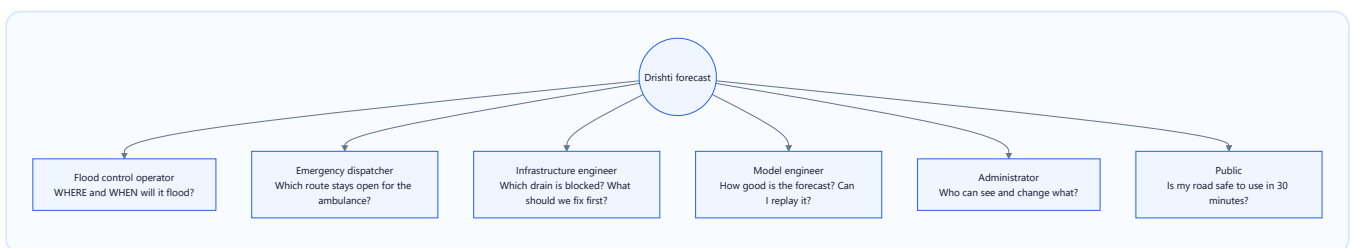
2. The problem, explained

2.1 Why urban floods are hard to predict



Say: "Flooding on a street is the result of three systems interacting: the rain, the ground, and the drains. You can't predict it from rainfall alone — our own baseline that uses only a rainfall threshold gets a CSI of 0.11 and nine false alarms out of ten. You need all three, coupled."

2.2 Who needs the answer



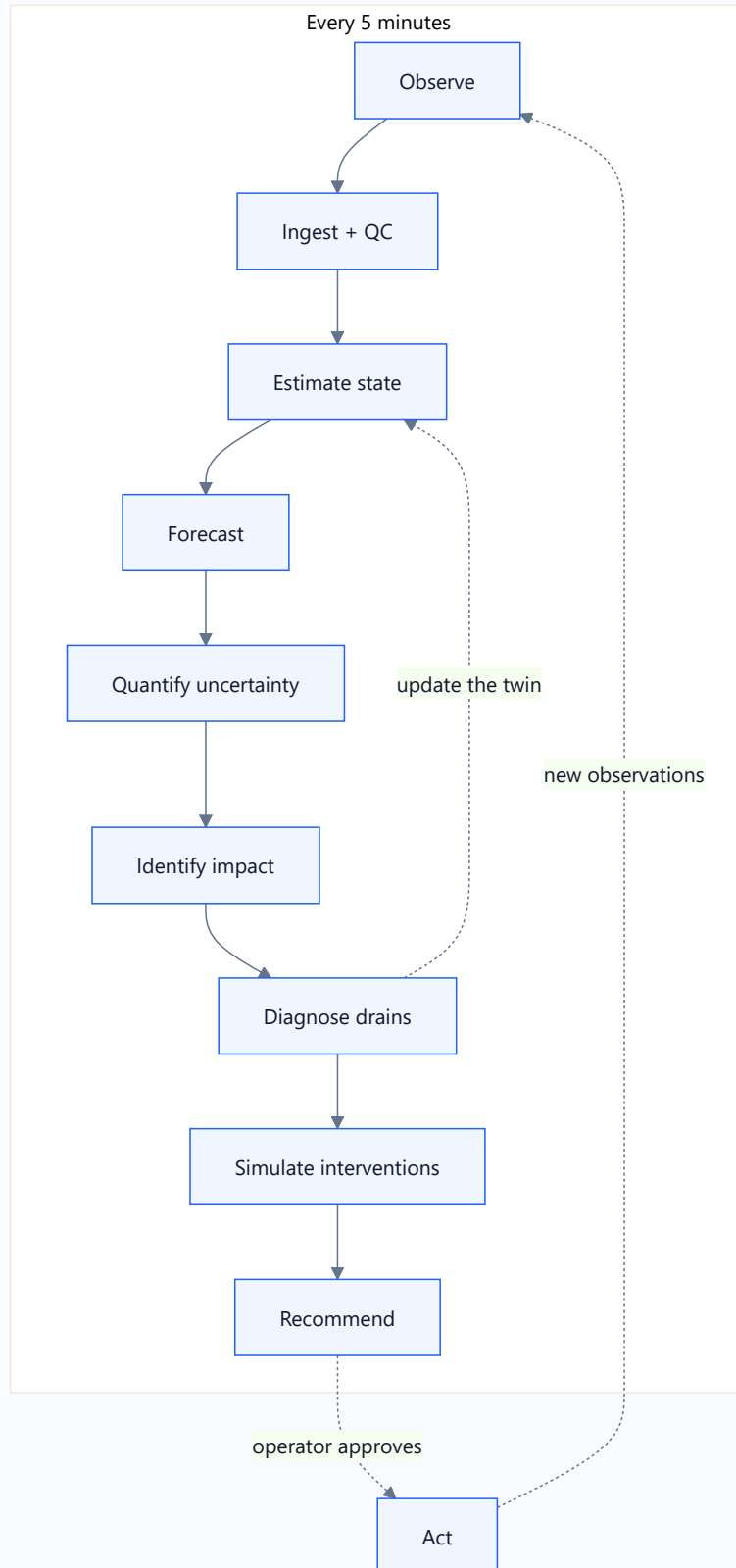
These six roles are real in the dashboard: the **role selector** in the header locks views per role (for example, a public viewer only sees Situation, Forecast and Roads).

2.3 The five questions

#	Question	Drishti's answer	Where it is shown
Q1	WHERE?	Flood depth on a 5 m grid, per road segment	Forecast view (B), map
Q2	WHEN?	Time to 10 cm, peak time, duration	Click any cell in view B
Q3	HOW BAD?	Expected / P10 / P50 / P90 depth, P(depth > 10/20/30 cm)	Views B and C
Q4	WHY?	Rainfall / runoff / terrain / drainage / asset / boundary factors	"Why here?" panel in view B
Q5	WHAT NOW?	Ranked interventions, 3 alternative routes, work orders	Views E and F

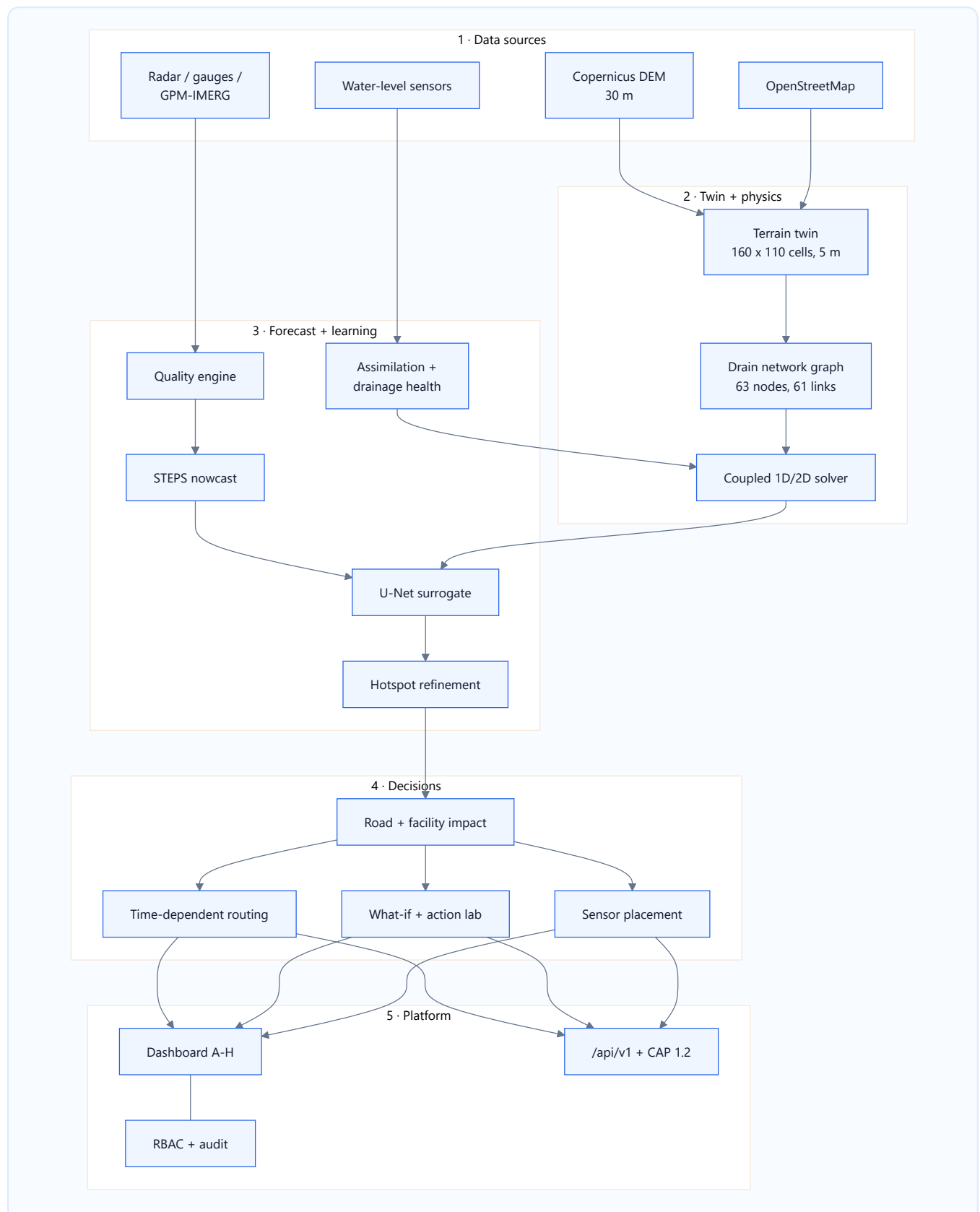
3. The solution, explained

3.1 The closed loop



Say: "This is the difference between a dashboard and a digital twin. A dashboard shows you data. A twin keeps a running model of the physical system, corrects it with every new observation, and uses it to look ahead."

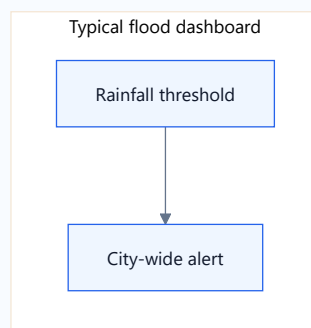
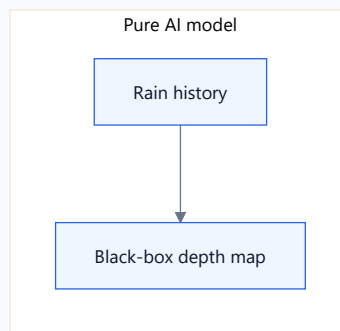
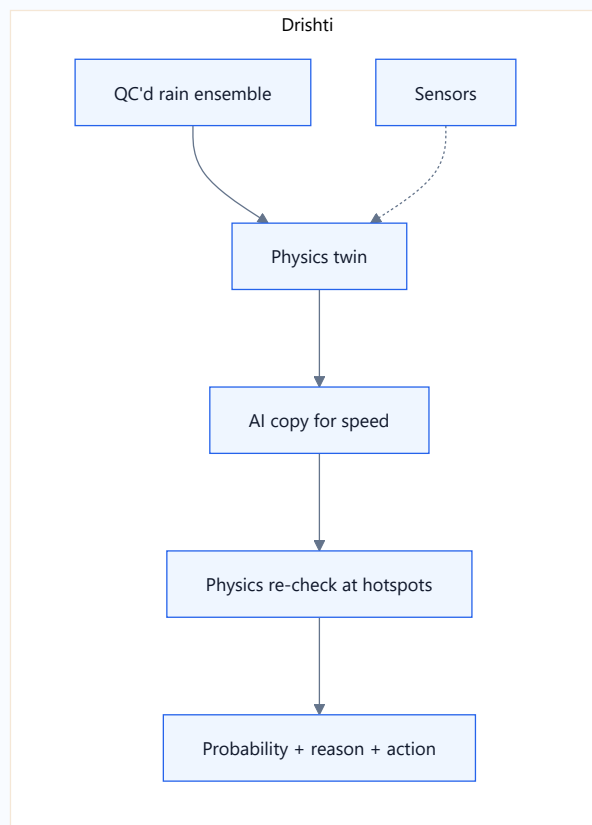
3.2 System architecture



3.3 The tech inside each box

Box	Method	Code
Terrain twin	Copernicus GLO-30 → bilinear to 5 m, priority-flood fill, D8 flow, low points; OSM roads and buildings rasterised	<code>simulation/terrain/twin.py</code>
Drain network	Inlets placed by flow accumulation + low points + road proximity; Dijkstra tree to outfalls; Rational-method sizing	<code>simulation/drainage/network.py</code>
Coupled physics	2D diffusive-wave storage cells + 1D local-inertial pipes + weir/orifice exchange	<code>simulation/hydraulics/simulate.py</code>
Quality engine	Range, spike, frozen, drift, stale, coverage, gauge-vs-radar checks	<code>forecast/quality.py</code>
STEPS nowcast	Optical flow + 6-level FFT cascade with AR(2) + spectral noise + NWP blend	<code>forecast/steps.py</code> , <code>forecast/motion.py</code>
Surrogate	Residual U-Net, 36 input channels → depth at 14 leads, exported to ONNX	<code>forecast/surrogate/</code>
Hotspots	80 m tiles; full physics for the P50 and P90 members where risk or spread is high	<code>forecast/adaptive.py</code>
Probabilistic forecast	Per cell and per road: P10/P50/P90, peak, time-to-10 cm, duration, exceedance	<code>forecast/probabilistic.py</code>
Assimilation + health	Per-sensor Kalman bias update, innovation-based fault isolation, residual z-scores	<code>tools/export_dashboard.py</code>
Routing	Time-dependent A* — each edge is costed at the predicted arrival time	<code>dashboard/app.js</code> , <code>api/route.py</code>

3.4 What makes it different



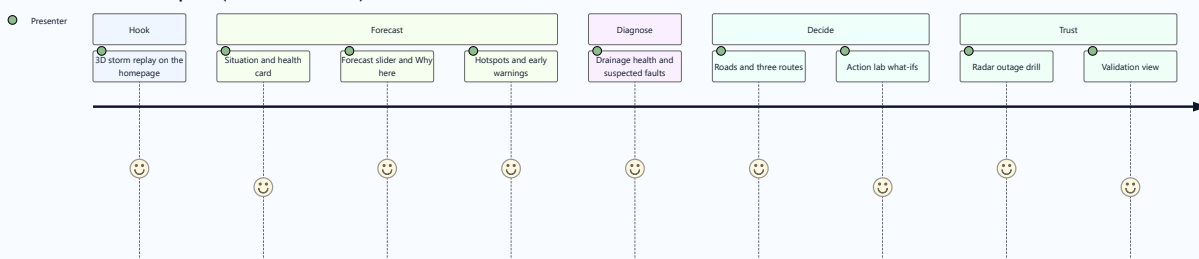
4. Live demo walkthrough

4.0 Setup (do this before you walk in)

```
cd Drishti
python -m http.server 8123
# open two tabs:
#   http://localhost:8123/                (landing page, 3D replay)
#   http://localhost:8123/dashboard/      (operator dashboard)
```

Backup: the same pages are hosted at <https://drishti-indol.vercel.app/> and <https://drishti-indol.vercel.app/dashboard/>. The 4:51 explainer video is https://youtu.be/MfLX_qhArzU (local copy: `media/drishti_explainer.mp4`).

The demo path (about 6 minutes)



Step 1 — The hook: the 3D replay (30 s)



Do: open the landing page and let the storm play. Drag to orbit.

Say: "This is the real KIET campus terrain, the real buildings from OpenStreetMap, and our drain network in green. The rain falling is a simulated squall moving across campus. Watch the water collect — it doesn't spread evenly, it runs to the low points and backs up where the drains can't take it. The water columns are drawn twelve times taller than life so you can see them. The panel on the right is live: rain rate, flooded area, the deepest street cell."

Do: click **Open the operations dashboard**.

Step 2 — Situation (view A) (45 s)



Point at: the **Forecast health** card (top right of the map, always visible), the header's **Forecast issued** selector and the green **OPERATIONAL · radar** pill.

Say: "This is what an operator sees first. Top right is the forecast health card — rain source, how many gauges are valid, how many sensors we trust, the uncertainty level and when the forecast was issued. It never leaves the screen, because a forecast is only as good as the data behind it. The header lets me replay three forecast cycles of the same storm: before it arrived, once it was established, and later."

Step 3 — Forecast (view B) and "Why here?" (90 s)



Do: press **B**. Drag the lead-time slider from +5 to +180 min. Switch **Expected / P10 / P50 / P90 / Peak**. Then click a coloured cell near a building.

Say: "Now I'm looking ahead. This slider runs from five minutes to three hours. Blue is shallow, orange is deep. These buttons switch between the expected depth and the pessimistic P90 — the depth we are 90% sure won't be exceeded. When I click a street cell I get its own forecast: depth now, depth at my chosen lead, expected peak and when, and **when it reaches 10 centimetres**. The grey lines are the twenty ensemble members — that spread is our uncertainty, shown, not hidden.

And underneath: **Why here?** Drishti breaks the risk into rainfall, runoff, terrain, drainage, known faults and the downstream boundary, and names the main driver. That answers the question every engineer asks: *why is this street flooding?*"

Step 4 — Hotspots (view C) (45 s)



Do: press **C**. Toggle > 10 / > 20 / > 30 cm. Point at the hotspot list and the radar mini-map.

Say: "This is the probability view — the chance each cell goes above 10, 20 or 30 centimetres. The outlined tiles are hotspots: places with high risk, high uncertainty, or next to a building. For those, Drishti re-runs the full physics model instead of trusting the AI copy. Some are marked **early warning**: still dry now, but more likely than not to cross 10 cm. At the forecast issued before the squall, these warnings came a median of 50 minutes ahead."

Step 5 — Drainage health (view D) (60 s)



Do: press **D**. Point at the **Assimilation** chart, the **Suspected faults** list and **Sensor trust**.

Say: "This is the part a normal flood map can't do. We hid two blocked pipes in the 'true' storm, and put nine virtual water-level sensors in the drains — one of them stuck. The chart compares the model running blind — the open loop — against the model corrected by sensors. Sensor-level error drops from 19.9 cm to 2.7 cm.

Then it diagnoses. Sensor S-27 has been reading the same value for 20 minutes while the model says the level is changing — trust drops to 0.28 and it is ignored. Pipe P-49 reads 30 cm higher than it should under the same rain — likely blocked, and it was one of the two we hid. Each fault comes with evidence, a likelihood and a work order — never a definitive claim, and never automatic control."

Step 6 — Roads & routes (view E) (60 s)



Do: press **E**. Change the vehicle class (car → ambulance). Pick **From** and **To** blocks, set **Leave: in 30 min**, press **Find three routes**. Toggle the **emergency policy**.

Say: "Floods matter because of what they cut off. Every road segment gets a status — passable, degraded, high risk, impassable — and the thresholds depend on the vehicle: a two-wheeler is in trouble at 7 cm, a fire truck can go through 35. The route planner is time-dependent: if I leave in 30 minutes, each road is costed at the water level **when I'm predicted to reach it**, not now. I get three routes with ETA and the probability of hitting deep water. Notice we never call a route 'safe' — only lower risk."

Step 7 — Action lab (view F) (45 s)



Do: press **F**. Pick **Pump P-60 fails**, tick **show change vs baseline**. Then point at the **Interventions** table and **Request approval**.

Say: "What if? Each of these is a full physics run of the same storm. If the pump fails, flooded area goes from 5,800 to 8,000 square metres and 3,400 cubic metres overflow from the drains. If the receiving nala backs up by two metres, flooded area doubles. Below, the interventions — run the pump, jet the drains, or both — ranked by effect and cost. The recommendation needs a human to approve it. And at the bottom: where to put the next sensor to cut uncertainty the most."

Step 8 — The honesty moment: radar outage drill (30 s)



Do: tick **Radar outage drill** in the header.

Say: "Now the radar goes down. This isn't a fake banner — it switches to a real forecast computed with no radar and a frozen gauge. Drishti falls back to the gauges, rejects the frozen one, widens the uncertainty and says **DEGRADED** at the top of every view. Skill drops — early-warning hits fall from 67% to 26% — but there are still zero false alarms, and the operator knows exactly how far to trust it."

Do: untick the drill.

Step 9 — Validation (view G) (30 s)



Say: "Finally, how good is it? Top: which parts of the model matter — rainfall alone scores 0.11, adding terrain 0.43, runoff 0.53, the full coupled model is the reference. Middle: the AI surrogate against persistence — at three hours it scores 0.79 where persistence falls to 0.24. Bottom: the forecast issued at T+115 — 67% of later-flooded cells warned in time, 2% false alarms. One honest caveat on screen: these are scored against our synthetic truth, not observed floods yet."

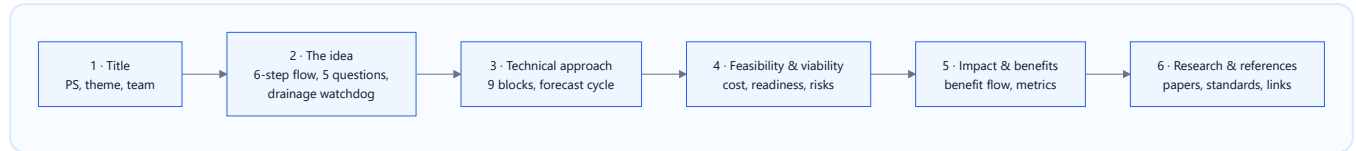
Optional — System (view H)



Say: "For the IT team: the five-minute loop and how long each stage takes, the API endpoints with their real responses, CAP 1.2 alert export for India's alerting systems, GeoJSON for navigation apps, the role matrix and the audit log."

5. Slide-by-slide talk track

The deck is `ppt/Drishti_SIH26085.pptx` (6 slides, SIH 2026 template). Speaker notes are embedded in each slide.



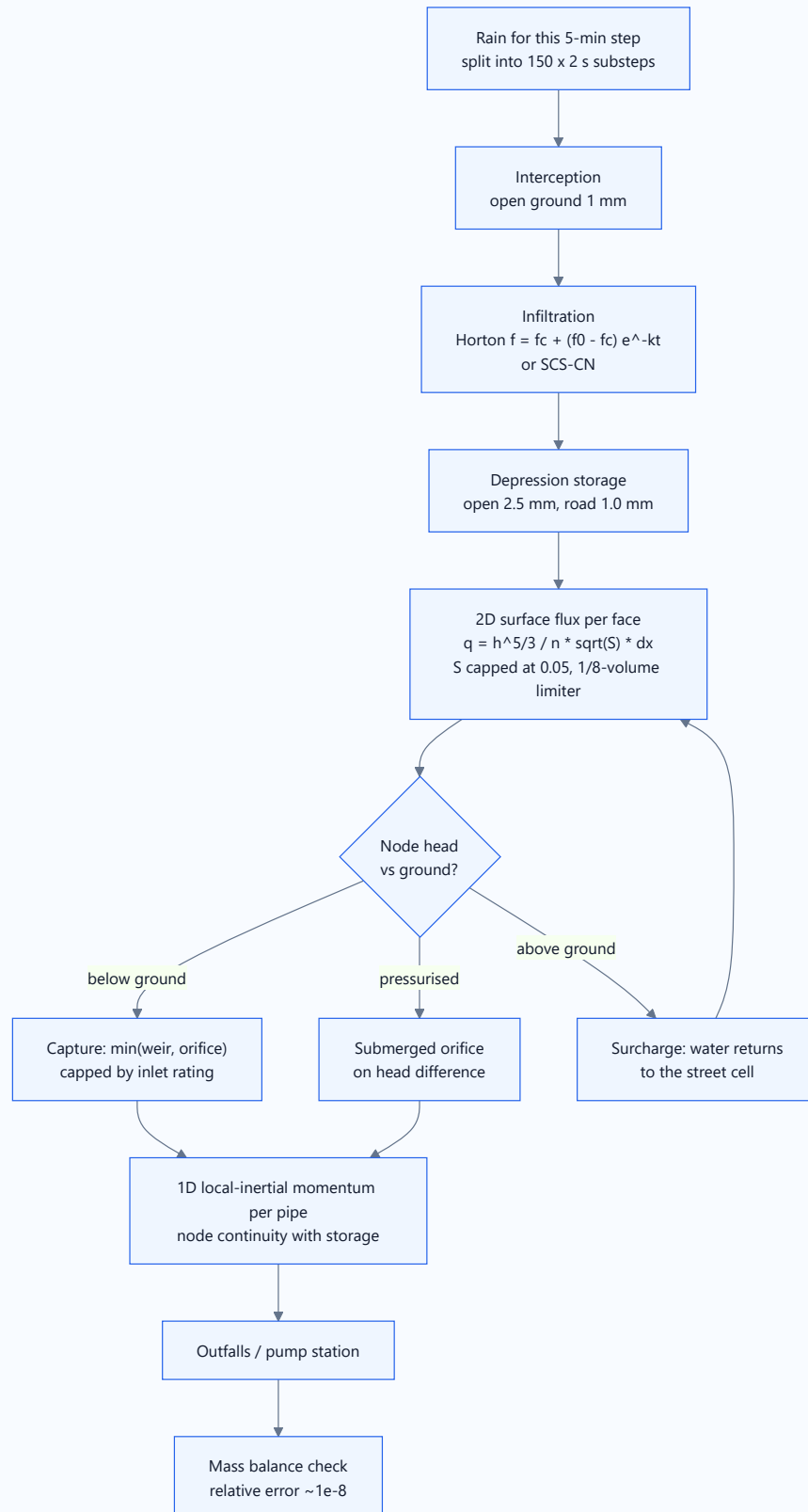
Slide	Time	Key line	Point at
1 · Title	15 s	"Drishti tells a city where water will stand on its streets, up to three hours before it does."	PS ID and title
2 · Idea	60 s	"Observe, check, forecast twenty futures, run physics, run the AI copy, decide — every five minutes."	The six cards left → right, then the five-question ladder
3 · Technical	90 s	"Physics where it matters, AI where it's fast."	Block 4 (the cycle) top to bottom, then the <i>Why hybrid?</i> box
4 · Feasibility	45 s	"It runs today — 19 seconds, on a laptop, with open data."	Readiness bars, the rainfall chart, the challenge table
5 · Impact	45 s	"Warnings before the water, routes for the ambulance, the right pipe fixed first."	The benefit flow row
6 · References	15 s	"Every method traces to published work; the links open the live system."	The app thumbnails

Slide 3 in detail — walk the centre column:

1. *Rain observations* → *Quality check* ("a faulty reading is rejected, and logged").
2. *Radar usable?* → if not, "gauges only, and we say DEGRADED".
3. *Ensemble nowcast* → "20 possible rain futures".
4. *Twin state at t_0* → *AI surrogate depths* → "0.2 seconds for all 20".
5. *Hotspot risk?* → "full physics refine only where it matters".
6. *Probabilistic forecast* → *Crosses 10 cm?* → early warning.
7. *Impact · routes · alert* → audit ledger.

6. Technical deep dive

6.1 One physics time step



Key equations

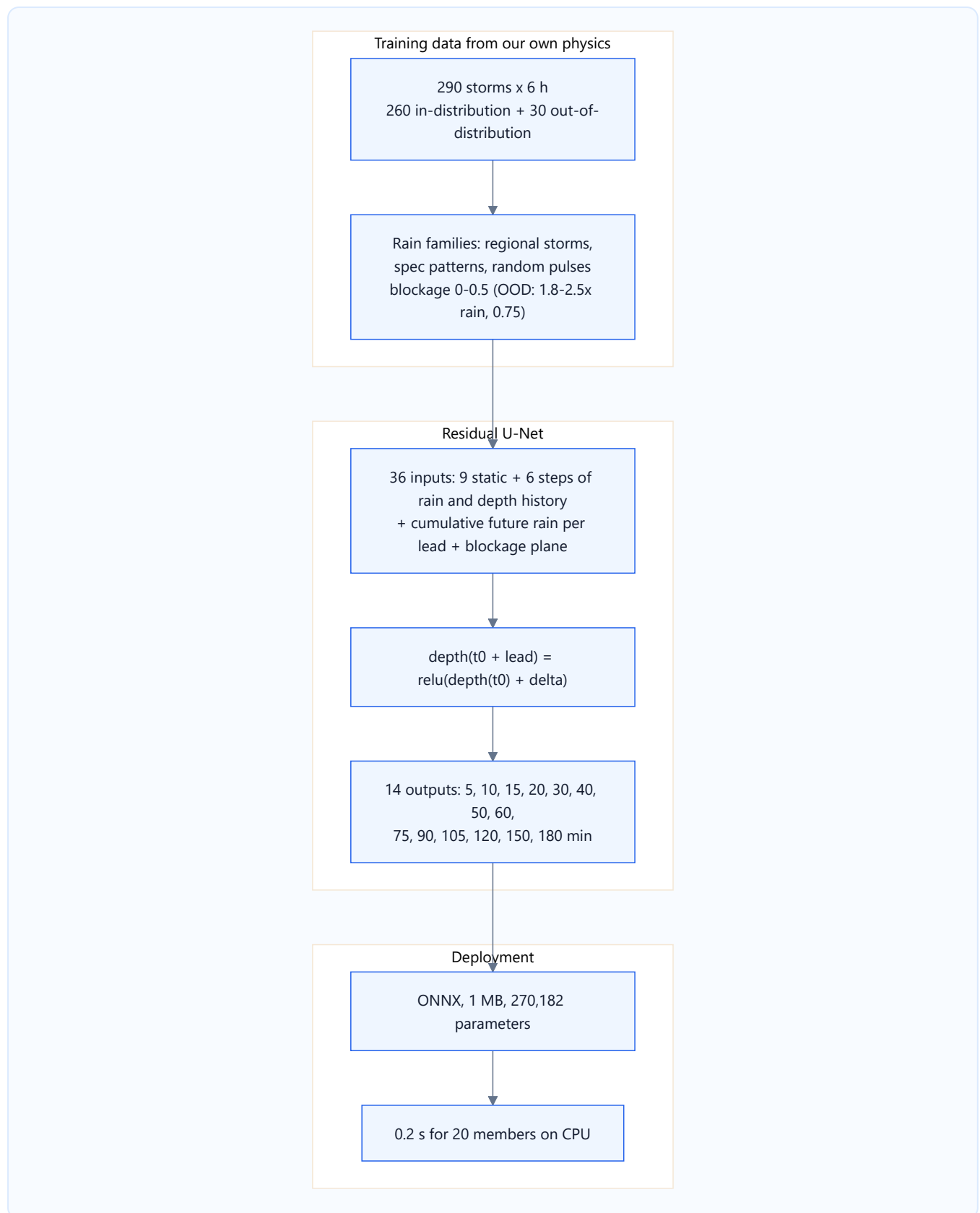
Process	Equation	Parameters
Surface flux (Manning, wide channel)	$q = h_{\text{flow}}^{(5/3)} / n \cdot \sqrt{S} \cdot dx$	$\sqrt{S} = \min($
Stability limiter	$q_{\text{face}} \leq 0.125 \cdot h_{\text{upwind}} \cdot dx^2 / dt$	conservative, explicit
Horton infiltration	$f(t) = f_c + (f_0 - f_c) \cdot e^{(-k t)}$	road 5/1 mm/h, open 35/8 mm/h
SCS-CN	$Q = (P - I_a)^2 / (P - I_a + S), S = 25400/CN - 254 \text{ mm}$	chosen in config/hydraulics.yaml
Pipe full capacity	$Q_{\text{cap}} = (1/n) \cdot (\pi D^2/4) \cdot (D/4)^{(2/3)} \cdot \sqrt{S}$	$n = 0.013$ concrete
Inlet exchange	weir $C_w \cdot P \cdot h^{1.5}$, orifice $C_o \cdot A_o \cdot \sqrt{2 g h}$	min of the two, capped
Mass balance	rain + inflow = infiltration + outflow + surface + depression + interception + node storage	reported per run

6.2 Rain nowcast (STEPS)



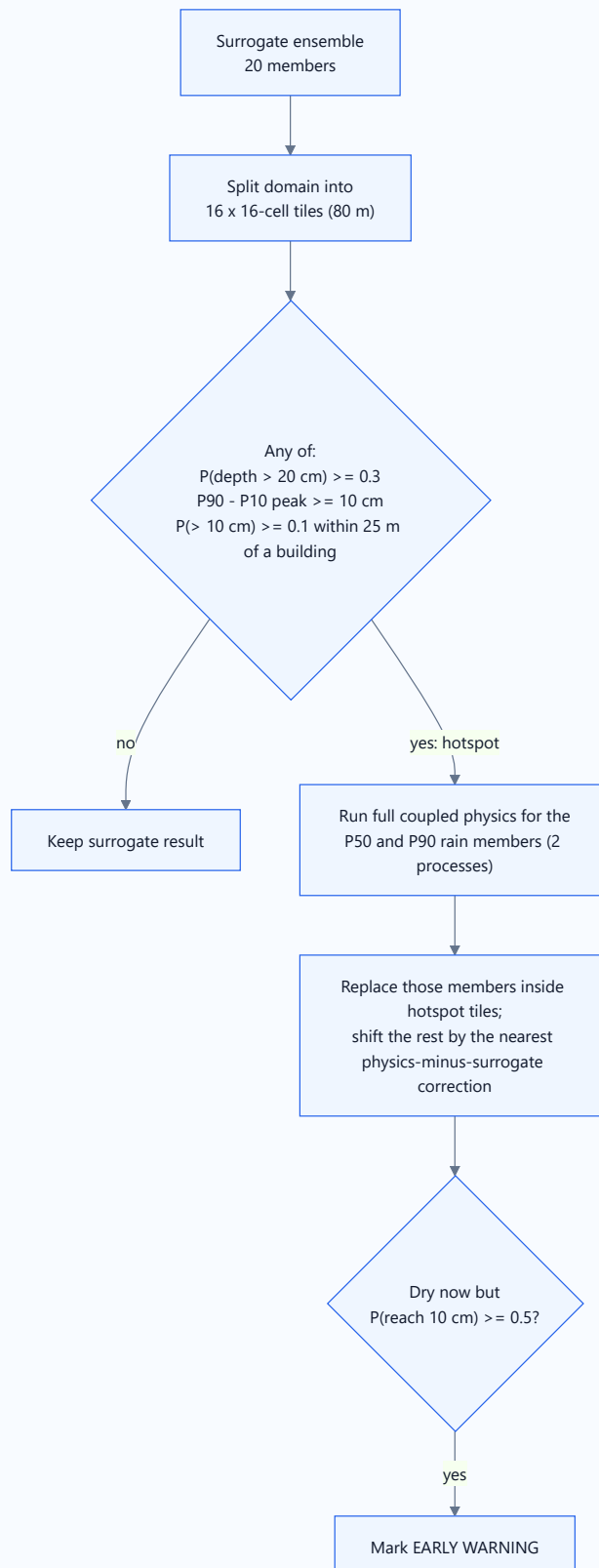
pysteps has no Python 3.14 wheel, so the algorithm is re-implemented in NumPy following pysteps. Member 0 is the unperturbed control.

6.3 AI surrogate: data → model → deployment

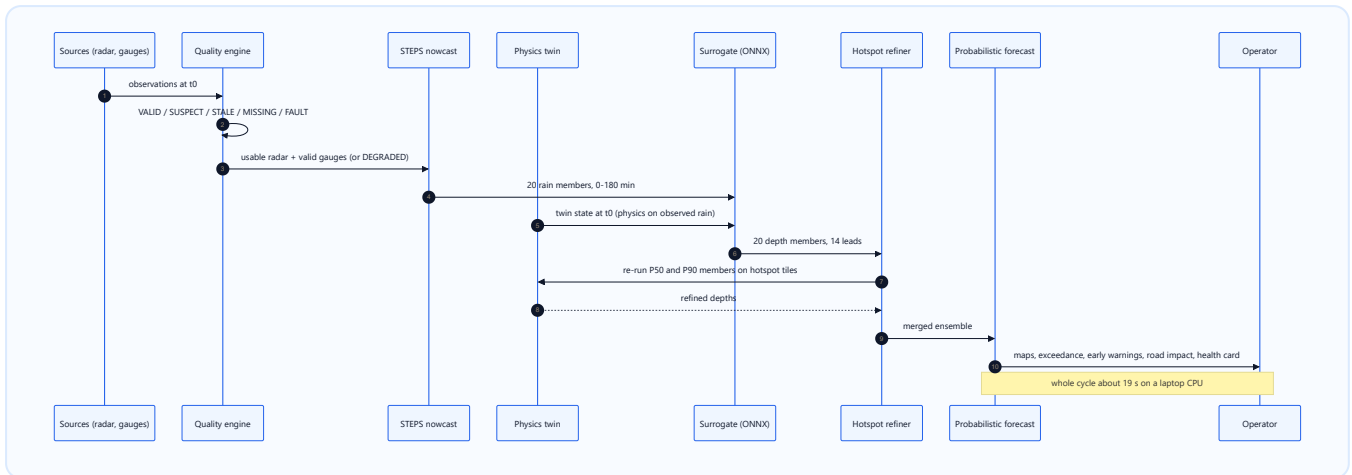


Splits are **by scenario, never by window**, so the test set contains storms the model has never seen.

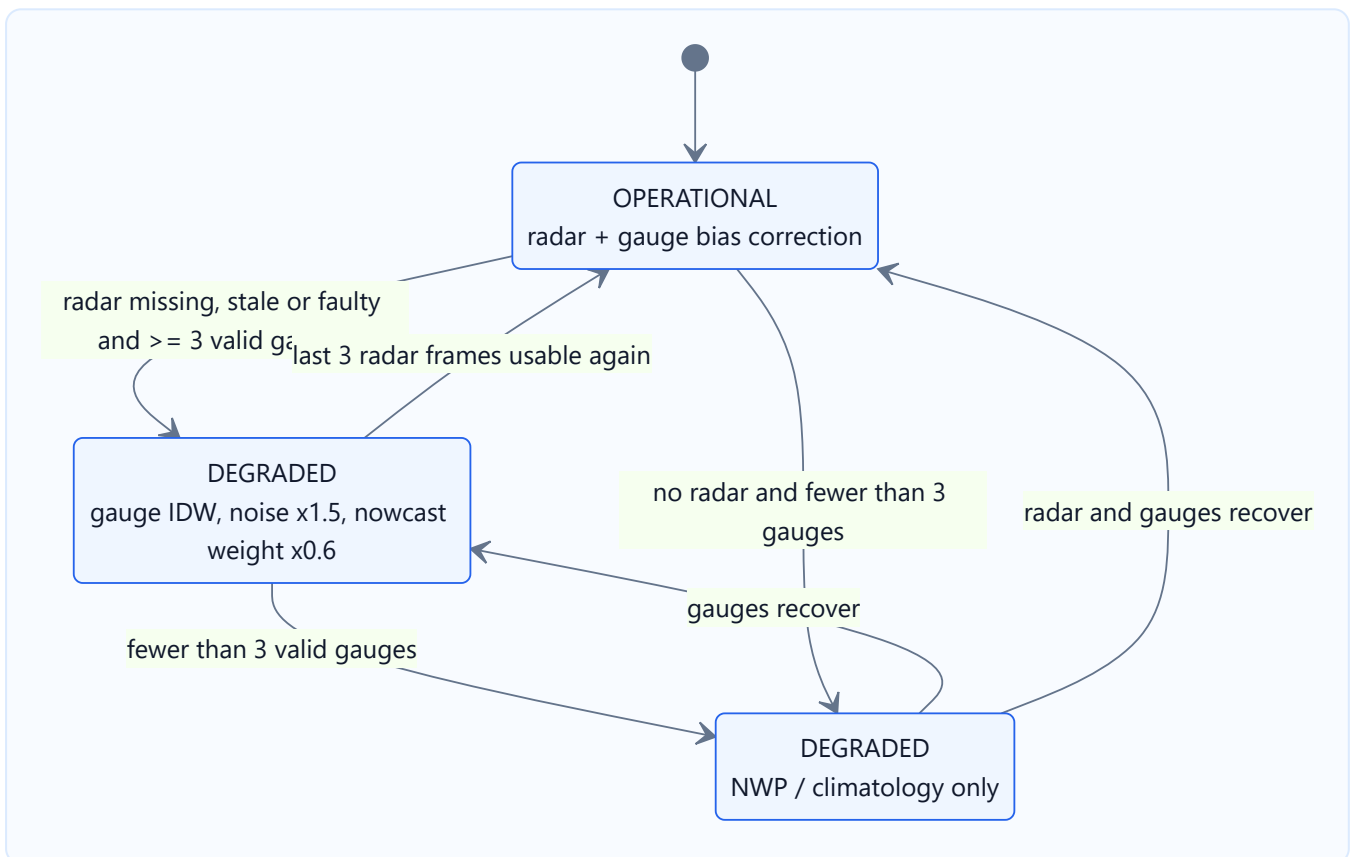
6.4 Adaptive fidelity (hotspot refinement)



6.5 Forecast cycle as a sequence

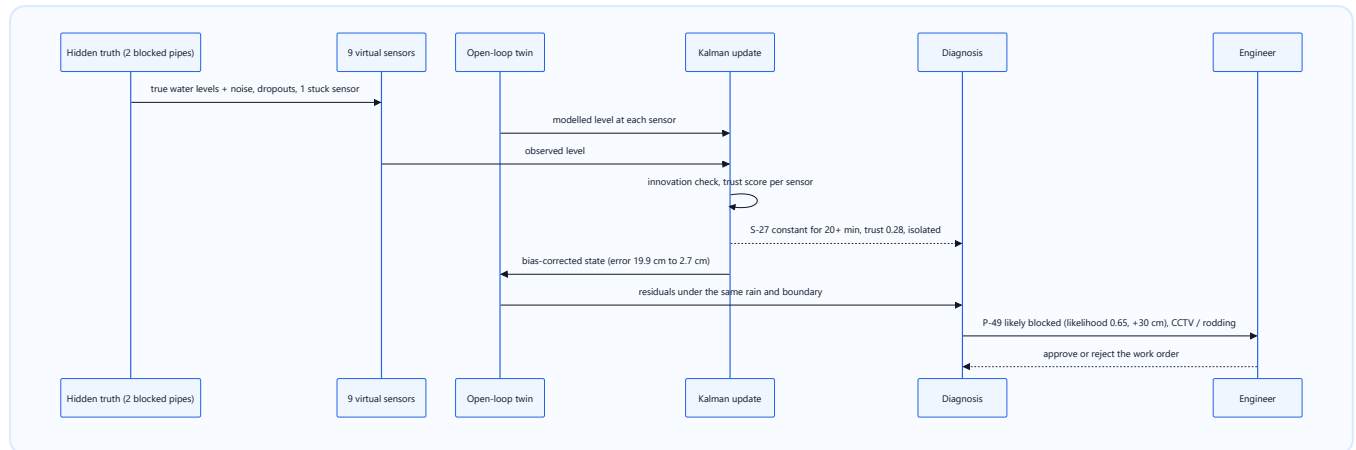


6.6 Degraded modes

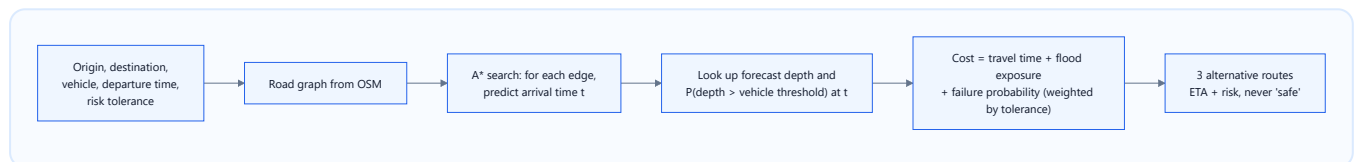


Every cause is listed in `health.reasons` (e.g. `radar MISSING`, `G3 = FAULT` (frozen)).

6.7 Closed loop: assimilation and diagnosis



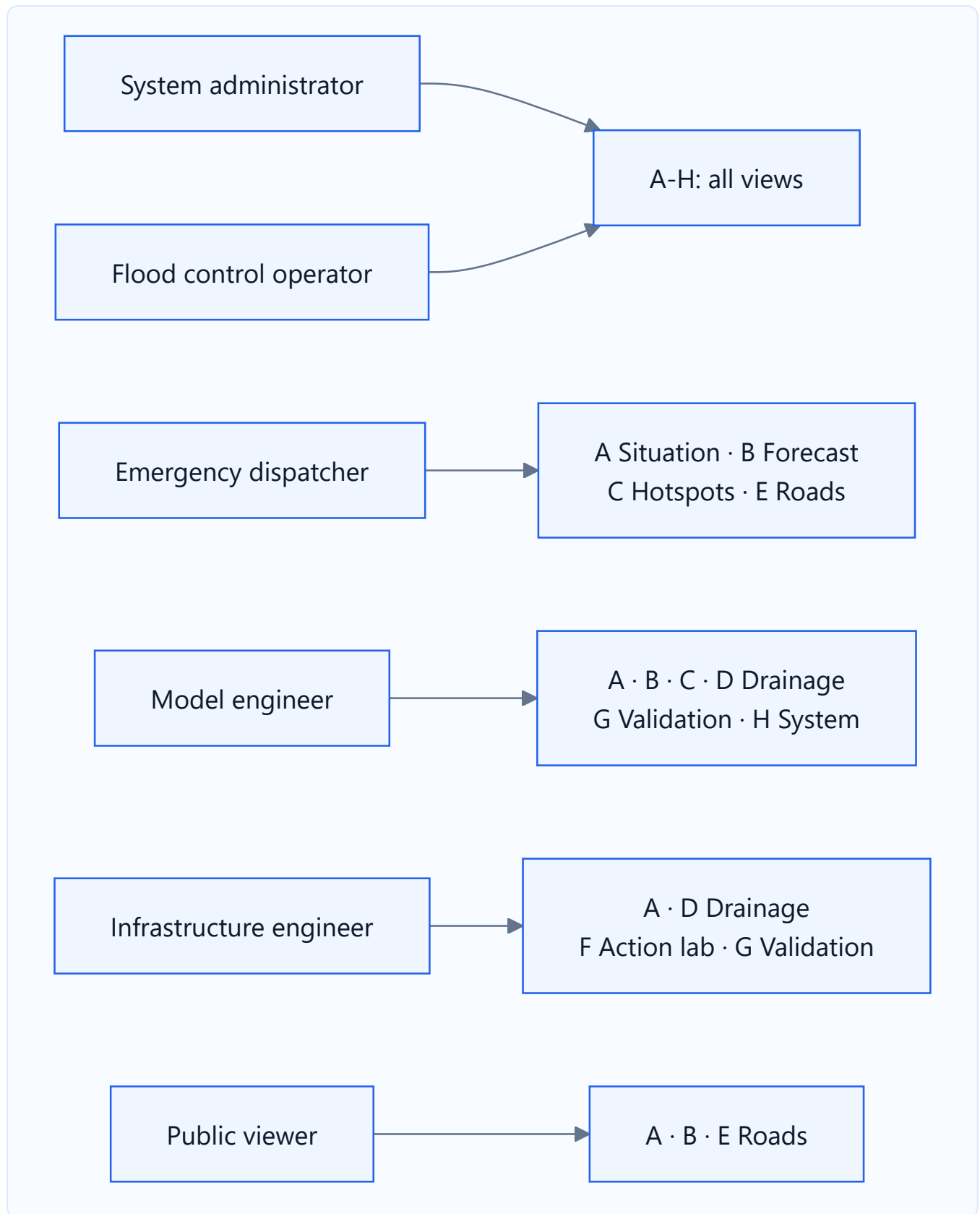
6.8 Routing



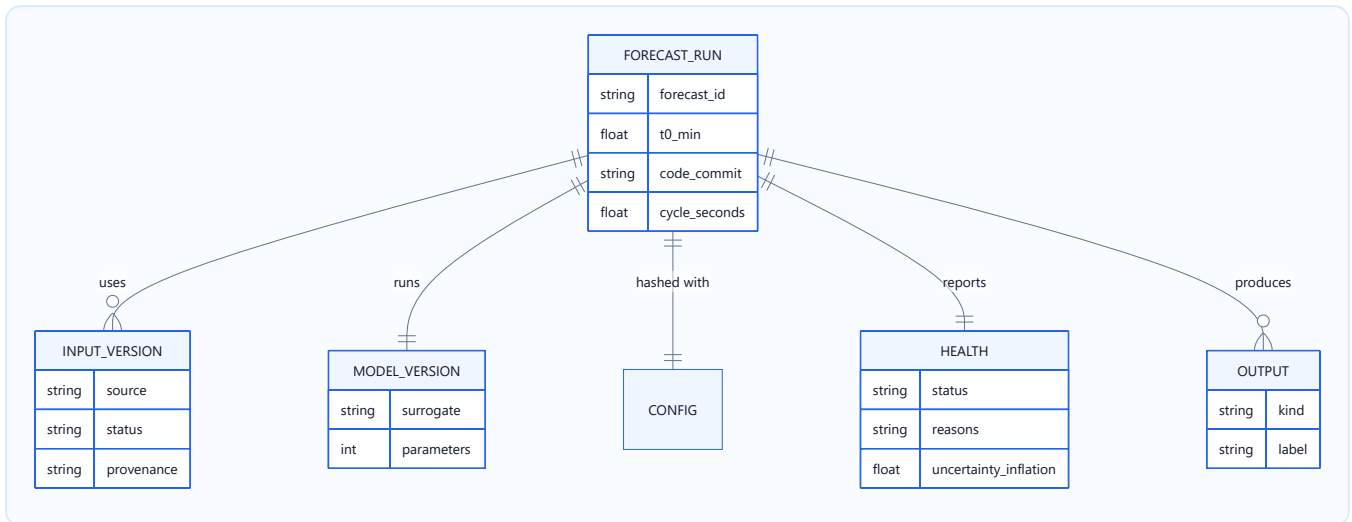
Vehicle	Degraded above	Impassable above
Two-wheeler	7 cm	15 cm
Car	10 cm	30 cm
Ambulance	15 cm	35 cm
Bus	25 cm	45 cm
Fire truck	35 cm	60 cm

The **emergency policy** toggle lets emergency vehicles use 5–10 cm deeper water and restricts public traffic earlier.

6.9 Roles and access

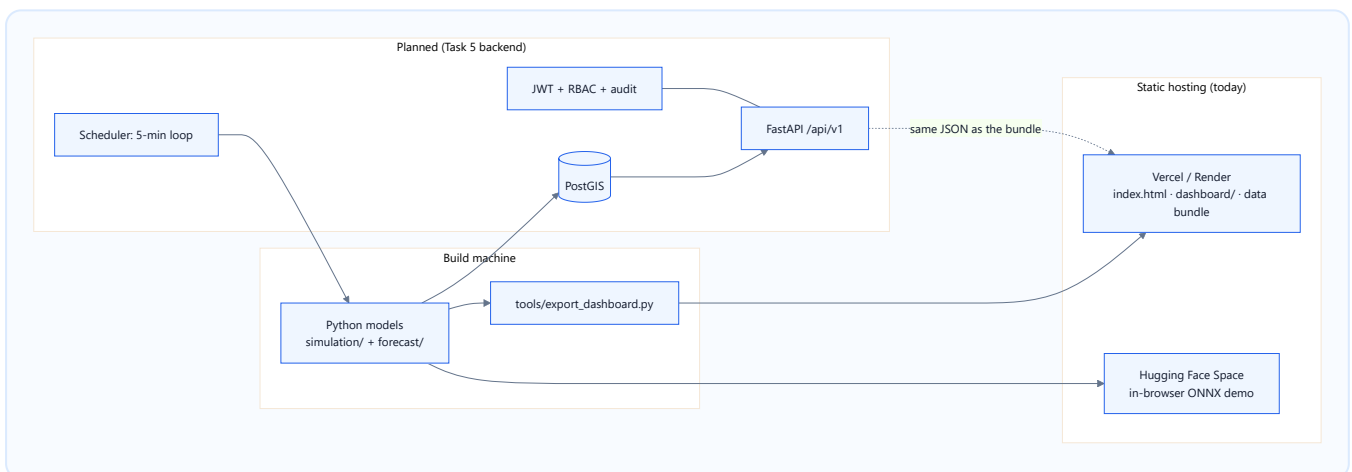


6.10 Provenance: every run is replayable



`label` is always one of *observed* / *derived* / *modelled* / *synthetic*, and the dashboard shows it as a tag on every panel.

6.11 Deployment



7. Q&A bank

Likely question	Answer
Are these real drains?	No. KIET has no surveyed drain map, so the network is generated from terrain and roads and every asset is tagged <code>verified=false</code> . A surveyed network drops in with the same JSON schema — no code change.
Can a 30 m DEM drive a 5 m model?	It gives the broad fall of the land, not kerb-level detail — and the UI says so (data tier 1). A LiDAR or drone DEM replaces it directly.
How do you know the model is right without flood data?	Three ways: the physics conserves mass to $\sim 1e-8$ and passes 31 tests; twin experiments score the forecast against a hidden simulated truth; and the replay/validation view is ready for observed events. We do not claim real-world skill yet.
Why not just train an AI on rainfall?	Our rainfall-threshold baseline scores CSI 0.11. Flooding depends on terrain and drains; an AI trained on our own physics learns that coupling, and hotspots are re-checked with the physics.
Why is the AI needed if you have physics?	20 members $\times \sim 12$ s of physics ≈ 4 minutes; the surrogate does all 20 in 0.2 s, so the cycle fits in 19 s.
How accurate is the surrogate?	CSI at 10 cm of 0.82 at +30 min and 0.79 at +180 min on held-out storms, vs 0.58 and 0.24 for persistence; it holds up on out-of-distribution storms (0.78 at +180).
What happens when the radar fails?	Gauge-only forecast, uncertainty $\times 1.5$, DEGRADED banner. In our drill, early-warning hits fell from 67% to 26% but false alarms stayed at 0%.
Are the probabilities calibrated?	Not yet — the ensemble is under-dispersive (P10–P90 covers the truth in 5–50% of active cells). Calibration needs assimilation of real data over many events; it is on the roadmap and stated on screen.
Does it control pumps automatically?	No. It recommends; a human approves. That is a design rule, not a missing feature.
How does it scale to a city?	The terrain twin, network and configs are per area; a ward is a config. The heavy work is offline training; the live loop is CPU-only.
What does it cost?	Indicatively ₹7–12.5 lakh for one ward, mostly sensors and a drain survey. Software and data are open; no GPU or per-forecast cloud bill.
What's not built yet?	The FastAPI server, database and authentication (the dashboard shows the exact responses each endpoint will return), and a full ensemble Kalman filter in place of the per-sensor update.
Which standards do you follow?	OASIS CAP 1.2 for alerts, GeoJSON for roads and routes, a SensorThings-style observation schema, EPA SWMM <code>.inp</code> export for cross-checking the drains.

8. Pre-demo checklist & fallbacks



- ☐ `python -m http.server 8123` running from the repo root
- ☐ Landing page and `dashboard/` open in two tabs; browser zoom 100%

- ☐ Forecast issued = **T+115 min**; Radar outage drill **off**; role = Flood Control Operator
 - ☐ Deep links ready: `dashboard/#forecast&lead=6, #prob, #drain, #impact, #actions&scen=pump_failure&diff=1, #live&drill=1, #valid`
 - ☐ `media/drishti_explainer.mp4` downloaded locally
 - ☐ Deck `ppt/Drishti_SIH26085.pptx` open in presenter view (speaker notes on)
-

9. Glossary

Term	Meaning
Nowcast	A forecast for the next 0–3 hours, built mostly from current observations
Ensemble / member	One of 20 equally plausible rain futures; their spread is the uncertainty
P10 / P50 / P90	Depth that 10 / 50 / 90% of members stay below
CSI	Critical Success Index: $\text{hits} \div (\text{hits} + \text{misses} + \text{false alarms})$; 1 is perfect
Persistence	The baseline "whatever is flooded now stays flooded"
Surrogate	A neural network trained to imitate the physics model, much faster
Surcharge	A drain so full that water is pushed back up onto the street
Backwater	Water in the receiving nala or river stopping the drains from emptying
Assimilation	Correcting the model's state with live sensor readings
Open loop	The model running without any sensor correction
STEPS	Short-Term Ensemble Prediction System — the rain nowcasting method
DEM / DSM	Digital elevation / surface model (a DSM includes buildings and trees)
CAP 1.2	Common Alerting Protocol — the standard format for public warnings
Level-1 data	Demonstration tier: public rain and terrain, synthetic drains and sensors